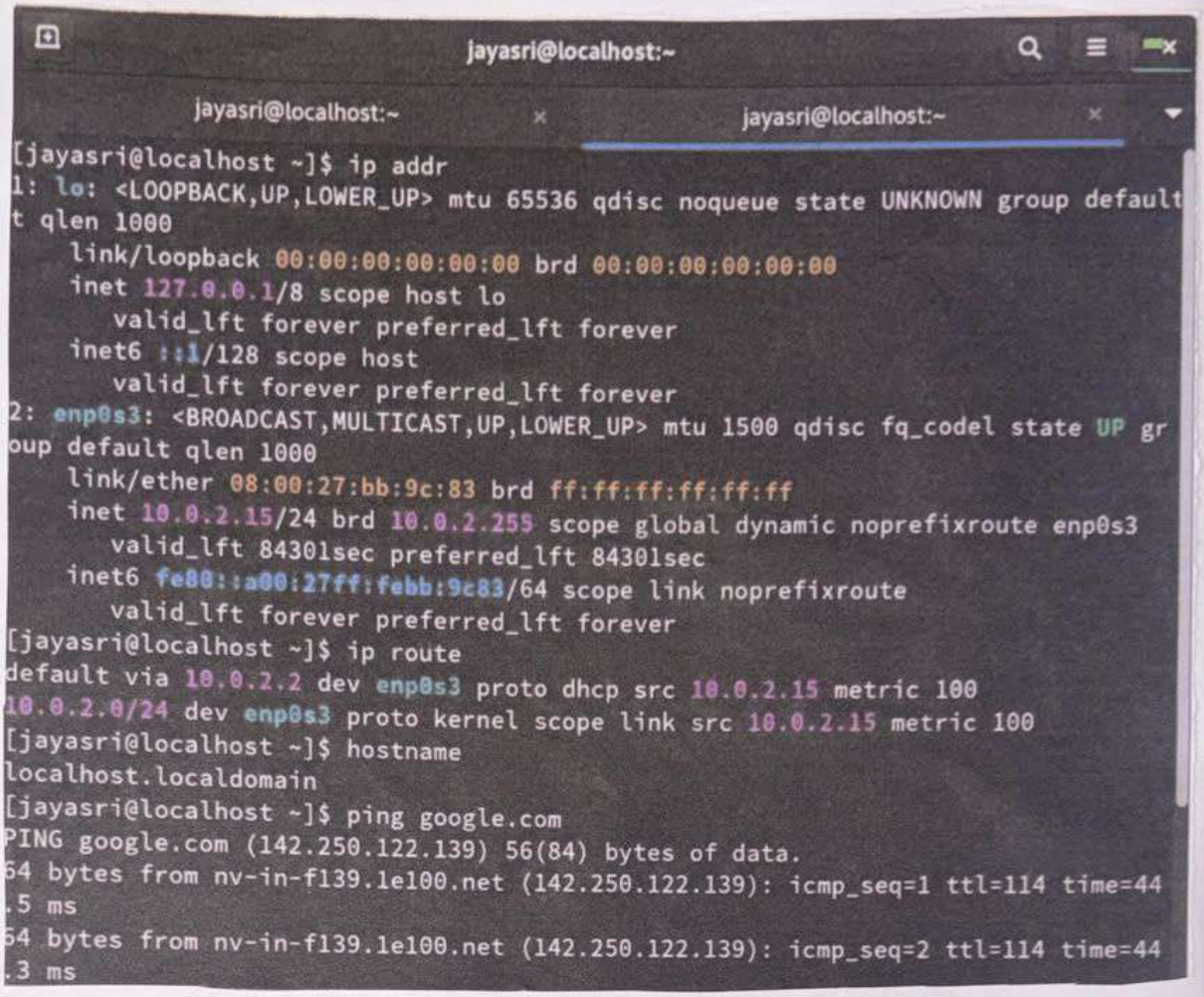
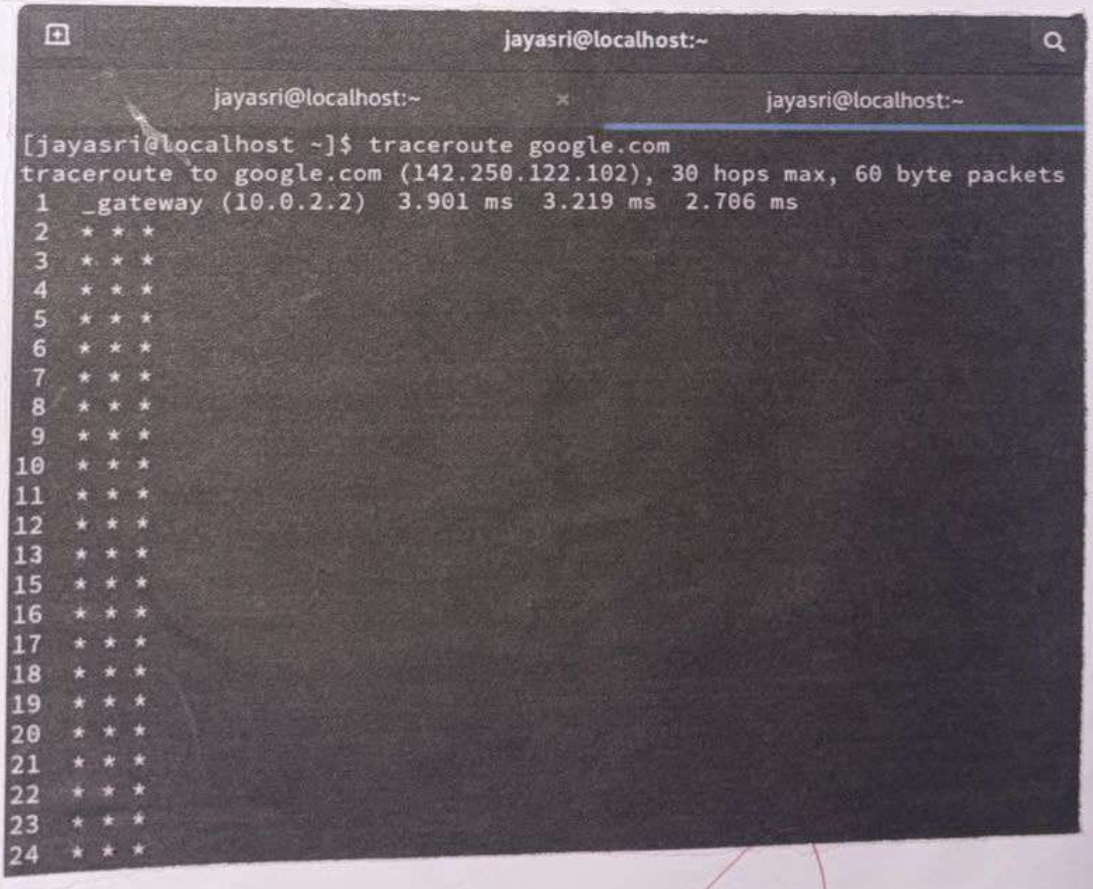

jayasri@localhost:~
jayasri@localhost:~
jayasri@localhost:~
[jayasri@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
t qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
link/ether 08:00:27:bb:9c:83 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
valid_lft 84301sec preferred_lft 84301sec
inet6 fe80::a00:27ff:febb:9c83/64 scope link noprefixroute
valid_lft forever preferred_lft forever
[jayasri@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[jayasri@localhost ~]$ hostname
localhost.localdomain
[jayasri@localhost ~]$ ping google.com
PING google.com (142.250.122.139) 56(84) bytes of data.
64 bytes from nv-in-f139.1e100.net (142.250.122.139): icmp_seq=1 ttl=114 time=44
.5 ms
64 bytes from nv-in-f139.1e100.net (142.250.122.139): icmp_seq=2 ttl=114 time=44
.3 ms


jayasri@localhost:~
jayasri@localhost:~
jayasri@localhost:~
[jayasri@localhost ~]$ traceroute google.com
traceroute to google.com (142.250.122.102), 30 hops max, 60 byte packets
1 _gateway (10.0.2.2) 3.901 ms 3.219 ms 2.706 ms
2 * * *
3 * * *
4 * * *
5 * * *
6 * * *
7 * * *
8 * * *
9 * * *
10 * * *
11 * * *
12 * * *
13 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 * * *
22 * * *
23 * * *
24 * * *

Exp no. 1
22/07/26

BASIC NETWORKING COMMANDS IN LINUX AND WINDOWS OPERATING SYSTEM

AIM:

To study and execute basic networking commands in windows and Linux operating system for viewing network configuration, testing connectivity, checking routine information, monitoring network status, and trouble shooting.

CODE:

Open Commands prompts

Commands:

```
ipconfig
ipconfig /all
Ping google.com
tracert google.com
nslookup google.com
route print
netstat
netstat -a
get mac
hostname
arp - a
```

Open Linux Networking Commands:

```
ip addr
if config
ip route
hostname
ping google.com
traceroute google.com
Sudo apt install traceroute
nslookup google.com
```

```
[jayasri@localhost ~]$ nslookup google.com
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.122.139
Name:   google.com
Address: 142.250.122.113
```

```
[jayasri@localhost ~]$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         _gateway        0.0.0.0         UG    100    0        0 enp0s3
10.0.2.0        0.0.0.0         255.255.255.0   U     100    0        0 enp0s3
[jayasri@localhost ~]$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost.localdo:38604 mirrors.ustc.edu.c:http TIME_WAIT
tcp        0      0 localhost.localdo:54522 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:38620 mirrors.ustc.edu.c:http TIME_WAIT
tcp        0      0 localhost.localdo:49892 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:54534 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:49904 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:38510 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:54512 101.6.15.130:http       TIME_WAIT
tcp        0      0 localhost.localdo:38608 mirrors.ustc.edu.c:http TIME_WAIT
tcp        0      0 localhost.localdo:38630 mirrors.ustc.edu.c:http TIME_WAIT
tcp        0      0 localhost.localdo:49918 101.6.15.130:http       TIME_WAIT
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ]         DGRAM      CONNECTED     22818    /run/chrony/chronyd.s
ock
unix  2      [ ]         DGRAM                    28072    /run/user/1000/system
d/notify
```

```
[jayasri@localhost ~]$ getmac
bash: getmac: command not found...
[jayasri@localhost ~]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[jayasri@localhost ~]$ ip neigh
10.0.2.2 dev enp0s3 lladdr 52:54:00:12:35:02 STALE
[jayasri@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:bb:9c:83 brd ff:ff:ff:ff:ff:ff
[jayasri@localhost ~]$
```

```
[jayasri@localhost ~]$ ip neigh
10.0.2.2 dev enp0s3 lladdr 52:54:00:12:35:02 STALE
[jayasri@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
de DEFAULT group default qlen 1000
    link/ether 08:00:27:bb:9c:83 brd ff:ff:ff:ff:ff:ff
[jayasri@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 192.168.1.1
[jayasri@localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
de DEFAULT group default qlen 1000
    link/ether 08:00:27:bb:9c:83 brd ff:ff:ff:ff:ff:ff
[jayasri@localhost ~]$ ss -tuin
Netid  State   Recv-Q   Send-Q        Local Address:Port      Peer Address:Port
udp    ESTAB   0        0          10.0.2.15%enp0s3:68           10.0.2.2:67
[jayasri@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
```

```
net stat - a
arp - a
ip link
cat /etc/ resource.conf
ip link show
ss - tucn
curl google.com

Sudo nmdi con mod "Wired connection 1"
        ipv4. getway 192.168.1.1
Sudo nmdi con mod "Wired Connection 1"
        ipv4. method manual
Sudo nmdi con mod "Wired Connection 1"
        ipv4. dns "8.8.8.8"
Sudo nmdi con up "Wired Connection 1"
```

RESULT:

Static IP address were successfully assigned to computers. Connectivity was verified using ping commands between hosts on the same subnet.